# Rotor Angle Stability and Inter-area Oscillation Damping Studies on the U.S. Eastern Interconnection (EI) under High Wind Conditions

Gefei Kou, *Student Member*, *IEEE*, Zhuohong Pan, *Member, IEEE*, Micah Till, *Student Member*, *IEEE*, Yilu Liu, *Fellow, IEEE*
University of Tennessee, Knoxville
Knoxville, TN, USA
gkou@utk.edu

Stanton Hadley, Tom King
Oak Ridge National Laboratory
Oak Ridge, TN, USA

***Abstract*—This paper studies the impact of wind generation on generator rotor angle and inter-area oscillation stability. A comparative simulation study is conducted based on a U.S. Eastern Interconnection (EI) planning model for year 2030. The wind penetration level is 17% of the total generation. Generator critical clearing time (CCT) and inter-area oscillation damping ratio are evaluated. It is observed that rotor angle stability is barely affected while inter-area oscillation damping shows discernible improvement due to wind generation. Possible explanations are then explored.**

***Index Terms*—Power system stability, power system simulation, power system model validation, wind power generation.**

## I. Introduction

Wind generation is gaining significant momentum as a new source of electric generation in the United States. Improved turbine technology and reduction in cost slashed the national average levelized price of wind Power Purchase Agreements (PPAs) to a historic low of 2.35 cents per kWh in 2014. With this competitive price, wind constituted 24% of electric generating capacity additions in the same year [1]. By the third quarter of 2015, the cumulative capacity of wind generation in the U.S. reached 69,471 MW [2] and is steadily moving towards the federal goal of 20% wind energy by 2030 [3].

Despite its economic and environmental competitiveness, the stability implications of wind generation must be studied with scrutiny. Type 3 and 4 wind turbines designs are decoupled from the grid side frequency and therefore are exempt of rotor angle stability issues. However, since wind generators do not contribute inherent inertia and have different voltage control mechanisms than their conventional counterparts, wind generation can introduce distinct boundary conditions for the remaining synchronous generators and thus change their rotor angle stability margin. Moreover, wind generation features rigid active and reactive power regulation. Without additional damping control functions, wind generators only lightly interact with the rest of the system. With the decommissioning and re-dispatch of conventional generators, system oscillation modes could experience dramatic change. The impact of wind generation on rotor angle and oscillation stability is therefore of interest for power system operation and planning and has been widely studied. By comparing generator rotor angle deviation in the New England 39-bus system, it is concluded in [4] that the displacement of synchronous generators with wind units could improve rotor angle stability. The improvement is directly influenced by reactive power control. A similar study is carried out on a 9-bus system in [5]. By comparing maximum angle separation and the transient stability index, it is discovered that wind generators with unity power factor and terminal voltage control modes can undermine rotor angle stability. As for power system oscillation, in [6] the contribution of wind generators to inter-area oscillation damping is studied on a 12 GW Southeastern Europe system with an instantaneous wind penetration level of 21%. Simulation indicates that double fed wind generators are contributing positive damping to inter-area modes. In [7] a 581 GW 22,000-bus system with a 0.8% wind penetration level is studied. By looking at the sensitivity of the eigenvalue with respect to inertia, the authors conclude that the damping ratio could either increase or decrease with additional wind generation. In [8] the oscillation modes are studied for a series of Western Electricity Coordinating Council (WECC) planning models in 2010, 2020, and 2022. The total generation is up to 178 GW. The converter control-based generators (CCBGs) make up 0.00%, 2.74%, and 20.22% of generation, respectively. It is found that CCBGs have low participation in the traditional inter-area modes but can introduce new modes. Those new modes may even be poorly damped without properly tuned wind control parameters.

The study documented in this paper is part of an effort to develop and analyze dynamic models for the U.S. Eastern Interconnection (EI) in year 2030. Previous efforts have covered dynamic model development [9], model validation [10], and frequency response [11]. This study looks at the effects of 17% wind penetration on the rotor angle stability of synchronous generators and inter-area oscillation damping. Different than previous studies on rotor angle and oscillation stabilities, this work used a realistic large-scale planning model, i.e., the 2030 EI dynamic models, for simulation. The detailed models include over 70,000 buses and 8,000 generators. The total generation capacity is 560 GW. The

The authors would like to acknowledge the support of the Oak Ridge National Laboratory and the U.S. Department of Energy's Office of Electricity Delivery and Energy Reliability under the Advanced Modeling Grid Research Program. This work was also supported in part by the Engineering Research Center Program of the National Science Foundation and the Department of Energy under NSF Award Number EEC-1041877 and the CURENT Industry Partnership Program.

978-1-5090-4168-8/16/$31.00 ©2016 IEEE

baseline model represents governor deadband and is validated against synchrophasor measurements for improved simulation accuracy. A large number of simulation cases are carried out to support the conclusions. This paper is organized into four sections. Section II introduces the studied dynamic models, the use of critical clearing time (CCT) as the rotor angle stability measure, and the oscillation computation algorithm. In Section III, simulation results are shown and summarized. Conclusions are provided in Section IV.

## II. Study Approach

### A. *Dynamic Modeling Approach*

The dynamic models used in this study were developed based on power flow models created by the Eastern Interconnection Planning Collaborative. This U.S. Department of Energy funded project models the impact of various policy options, technological advances, and economic factors on the power grid. Two power flow scenarios for year 2030, i.e., the High-wind and the Business-as-usual scenarios, were further examined by the research team at the University of Tennessee (UTK) and the Oak Ridge National Laboratory (ORNL). To analyze the stability impact of high wind penetration, the dynamic models were created. The dynamic modeling process involved parameterization of dynamic device models, a dynamic simulation initialization check, and N-1 dynamic contingency simulation [9]. After the dynamic models were adjusted to be stable, actual events were duplicated to check the degree of correlation between simulation and measurement. The synchrophasor measurements used were collected by FNET/GridEye and have a sampling rate of ten points per second [12]. The system validation work indicated that the predicted EI frequency response was stronger than the true value, largely because a key parameter, governor deadband, was not standardly represented in interconnection-level models. By inserting and adjusting governor deadband, simulation could show improved alignment with true response [10, 11, 13, 14]. The simulation models used in this study were indirectly validated to measurement by displacing wind generators with conventional ones to resemble current system behavior. The details on model validation are documented in [10, 11].

The stability impact of wind generation is multi-folded. Higher wind penetration usually comes with an upgraded transmission network, which enhances system stability per se. Higher wind penetration also implies re-dispatch of power flow and de-commissioning and de-loading of conventional generators. This change of system operating condition translates into a shift of stability performance. Lastly, the control mechanisms and reactions to system disturbances differ dramatically between wind and synchronous generators. The scope of this study is limited to the latter aspect. The two comparison cases, i.e., the baseline case and the high-wind case, are built upon the power flow scenario with 17% instantaneous wind generation. The total generation is 560 GW. The majority of wind plants are located in the footprint of Southwest Power Pool and Midcontinent Independent System Operator. Except for the wind generator models, the two dynamic cases are identical. The high-wind case represents wind farms with the General Electric (GE) wind machine model, while the baseline case models the wind farms with paired synchronous generator and exciter models. Some units are also equipped with turbine governor and power system stabilizer models. The frequency response of the baseline case is calibrated against the current system by adjusting active governor ratio and governor deadband. Since the only difference between the two dynamic cases is the dynamic wind farm models, the difference in system dynamic behavior can be traced directly to the difference between wind and conventional generators.

There are 726 wind farms online, generating 95.42 GW in the initial operating condition. Wind farm locations are marked in Figure 1. Each wind farm is represented by a GE wind turbine generator (WTG) model [15]. With proper setup, this model can be used to represent either Type 3 doubly-fed asynchronous generators or Type 4 full converters. When properly parameterized, the GE wind WTG model can not only simulate GE-specific WTGs but also the generic WTGs, whose dynamic behaviors are completely dominated by converter controls. More generic and accurate WTG dynamic models have been developed, such as WECC second generation wind turbine models [16], and the refinement in WTG modeling will continue. The use of the GE WTG is valid because this model essentially captures the characteristics of converter-based generators, which have no inherent inertia and tightly regulated active and reactive power output. As the purpose of the impact study is to evaluate the most severe but realistic conditions, it is assumed that all WTGs have no additional active power control functionality like artificial inertia control and wind governor response. Reactive power regulation is set to the voltage control mode, in which the voltage of a designated bus is regulated to a pre-defined reference. One can expect that the ideal reaction of the WTGs to a disturbance is to keep active power unchanged and maintain the voltage magnitude of the terminal or a remote bus.

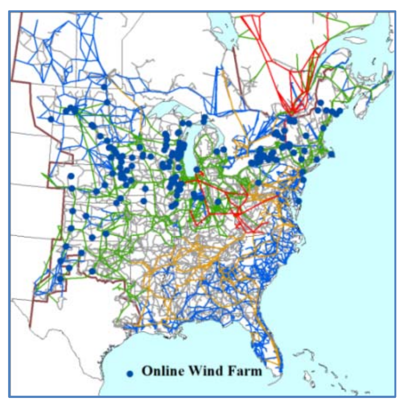



Figure 1. Online Wind Farm Locations in the Studied Cases

### B. *Critical Clearing Time Calculation*

Generator rotor angle stability is the ability of a synchronous generator to maintain synchronism when subjected to a severe disturbance. Instability is usually in the form of aperiodic angular separation due to insufficient synchronizing torque. The phenomenon is also referred to as first swing stability and usually occurs within the first three seconds following a disturbance [17]. The dynamics of rotor angle stability are depicted by Equations (1) and (2).

$$P_e = \frac{E_r E_I}{X_{eq}} sin\theta \tag{1}$$

$$P_m - P_e = \frac{H}{\pi f_0}\frac{d^2\delta}{dt^2} \tag{2}$$

Where:

978-1-5090-4168-8/16/$31.00 ©2016 IEEE

$P_e$: Electric power output $P_m$: Mechanical power input

$E_r$: Voltage magnitude on the receiving side

$E_I$: Generator internal voltage $X_{eq}$: Equivalent reactance

$\theta$: Power transfer angle $\delta$: Rotor angle

$H$: Generator turbine inertia $f_0$: Nominal frequency

When a fault occurs near a generator, its electric power output immediately drops due to depressed voltage magnitude. Since the mechanical power input on the generator shaft does not change during the first swing period, the imbalance between the input mechanical power and output electrical power accelerates the shaft speed and tends to break synchronism with the rest of the system. If the protection relays operate properly, the fault will be cleared quickly enough that the generator remains synchronized. However, if fault clearing is delayed, the generator could lose synchronism despite the action of protection relays. This happens because the post-fault power transfer capability may not be adequate to deliver the excessive energy accumulated on the shaft during the fault period. The maximum fault clearing time for a generator to remain synchronized is defined as the critical clearing time (CCT). CCT is a direct measure of a generator's rotor angle stability margin. CCT may change depending on initial system conditions, generator controls, and disturbance locations. It is also closely linked to boundary conditions. Stronger receiving end voltage support and smaller power transfer reactance can raise rotor angle stability margin and thus increase CCT.

To evaluate the rotor angle stability impact of wind generation, a standardized CCT calculation process was developed as follows. Most synchronous generators are usually connected to an external system through a single transformer or line. Then, a parallel branch is added. The impedances of the branch pair are kept equal and adjusted to be equivalent to the original branch so that power flow does not change. After applying a fault on one branch, the fault is cleared by tripping the faulted branch as shown in Figure 2. By examining the angle difference between the generator terminal bus and the external system bus, synchronism can be determined. Through repeated dynamic simulation, the CCT is obtained. Note that the CCT step used in this study is half a cycle.

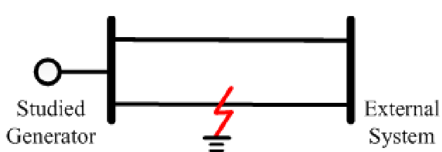



Figure 2. Branch Fault Setup

*C. Inter-area Oscillation Damping Calculation*

The matrix pencil method is used for estimating oscillation modes in simulation data. The matrix pencil method approximates an oscillatory signal by a summation of exponentially varying sinusoids [18]:

$$y(t) = \sum_i^n a_i \exp(b_i t) \cos(\omega_i t + \phi_i) \qquad (3)$$

Where:

$y(t)$: The oscillatory signal

$n$: The number of oscillation modes

$a_i$: The initial amplitude of oscillation mode $i$

$b_i$: The damping coefficient of oscillation mode $i$

$\omega_i$: The oscillation frequency of oscillation mode $i$

$\phi_i$ : The initial phase of oscillation mode $i$

In the matrix pencil method, a Hankel matrix is first formed from $y(t)$. Then, singular value decomposition (SVD) is used to extract singular values that are greater than a user-defined threshold. The remaining singular values are discarded. Last, by QR decomposition, a least square equation is solved to obtain $a_i$, $b_i$, $\omega_i$, and $\phi_i$.

## III. Simulation Results

This section documents the dynamic simulation results. Due to the volume of data and graphs, only sample cases are showed and a summary of the remaining cases is included.

*A. Rotor Angle Stability*

34 conventional generators are selected for rotor angle stability study. Those 34 units are geographically located across the system as shown in Figure 3. As previously mentioned, fictitious parallel branches are created on generator terminal buses. A fault is applied on a parallel branch to trigger a severe disturbance. By incrementally increasing the fault clearing time, the rotor angle of a subject generator will eventually diverge with the angle of the external system and the CCT is obtained. Table 1 summarizes the CCT comparison of the 34 generators between the baseline and high-wind cases.

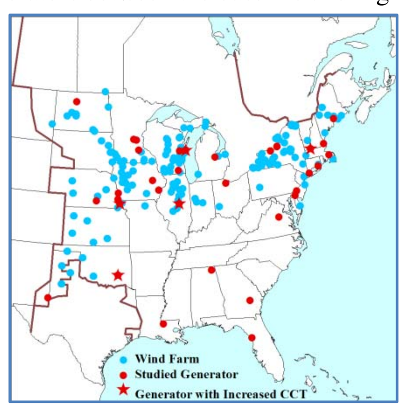



Figure 3. Location of Studied Synchronous Generators

Table 1. Comparison of CCTs (Unit: Half Cycle)

| Generator # | Baseline | High-wind | Generator # | Baseline | High-wind |
|---|---|---|---|---|---|
| 1 | 116 | 116 | 18 | 24 | 24 |
| 2 | 11 | 11 | 19 | 17 | 17 |
| 3 | 17 | 17 | 20 | 21 | 21 |
| 4 | 15 | 15 | 21 | 28 | 28 |
| 5 | 138 | 138 | 22 | 26 | 26 |
| 6 | 35 | 35 | 23 | 19 | 19 |
| **7** | **36** | **37** | 24 | 25 | 25 |
| 8 | 16 | 16 | 25 | 18 | 18 |
| 9 | 26 | 26 | **26** | **13** | **14** |
| **10** | **19** | **20** | 27 | 187 | 187 |
| 11 | 35 | 35 | 28 | 21 | 21 |
| 12 | 24 | 24 | 29 | 174 | 174 |
| 13 | 77 | 77 | 30 | 30 | 30 |
| 14 | 27 | 27 | 31 | 20 | 20 |
| 15 | 21 | 21 | 32 | 23 | 23 |

978-1-5090-4168-8/16/$31.00 ©2016 IEEE

| 16 | 17 | 18 | | 33 | 24 | 25 |
|---|---|---|---|---|---|---|
| 17 | 199 | 199 | | 34 | 203 | 203 |

It is observed that WTGs have no discernible impact on CCT in 29 out of 34 generators. For the remaining five generators, which are located in or near wind generation areas, WTGs increase CCT by a half cycle. Those generators are marked by a star in Figure 3.

To explore the physical explanation, an in-depth analysis is conducted on Case 33, shown in Figure 4. The generator rotor angle responses are plotted in Figure 5. The fault clearing time is the CCT of the high-wind case. As expected, the subject generator in the high-wind case stays synchronized but loses stability in the baseline case. Since CCT is directly tied to power transfer strength and boundary conditions, the difference is traced back to the contribution of WTGs. Figure 6 plots the reactive power responses of a nearby wind plant. In the high-wind case, the wind plant is represented by a WTG, while in the baseline case the same plant is modeled by a synchronous generator with excitation. The comparison indicates that the WTG provides faster and stronger reactive support than its conventional counterpart and thus increases transfer capability throughout the first swing.

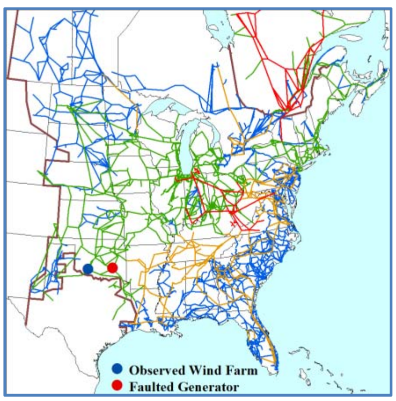



Figure 4. Generator Location for Case 33

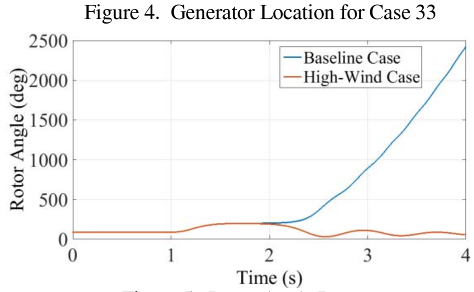



Figure 5. Rotor Angle Response

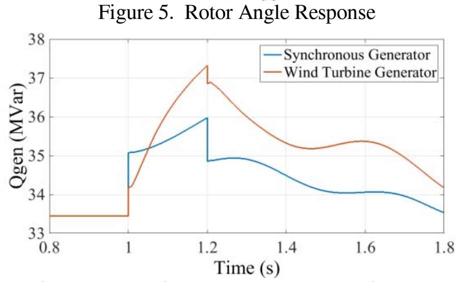



Figure 6. Reactive Power at a nearby Wind Plant

## B. *Inter-area Oscillation Damping*

To excite inter-area oscillation modes, 28 generators are tripped across the system. The matrix pencil method is used to calculate bus frequency oscillation modes at 16 observation points across the system. A sample case is plotted in Figure 7Figure 8. The tripped generator is located in the state of Georgia. The studied inter-area oscillation frequency is in the range of 0.15 to 0.20 Hz. The comparison of oscillation damping ratio is plotted in Figure 9.

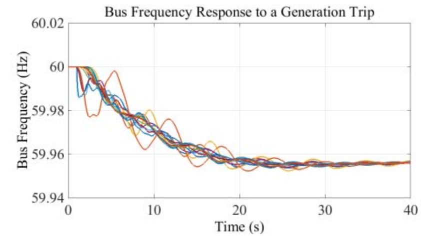



Figure 7. Frequency Response of the Baseline Case

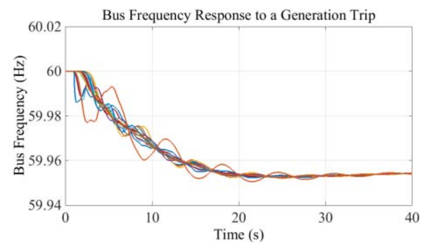



Figure 8. Frequency Response of the High-wind Case

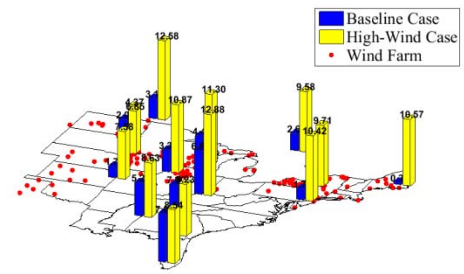
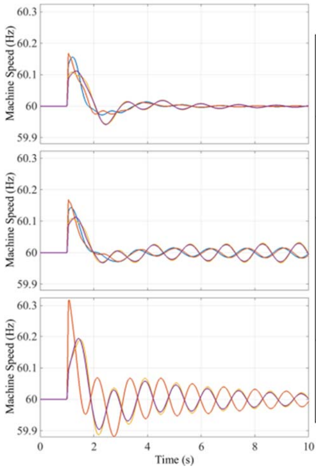



Figure 9. Damping Ratio Comparison

In this sample case, the high-wind case shows an average 434% increase in inter-area oscillation damping ratio, 1% increase in oscillation frequency, and 9% increase in oscillation magnitude. Due to the data volume, detailed results are not listed in full in this paper. By average the high-wind case increase the inter-area oscillation damping ratio by 183%, frequency by 7%, and magnitude by 9%.

It is physically intuitive that WTGs can increase inter-area oscillation frequency and magnitude due to the reduction of system inertia. To further confirm that WTGs' improvement in damping is always the case, a separate simulation test is conducted on the classical two-area system [19]. With the rest of the system kept identical, one of the four generators is modeled by: 1) High-gain excitation; 2) low-gain excitation with a power system stabilizer (PSS); and 3) a WTG in terminal voltage control mode, respectively. Figure 10 compares the simulation results and inter-area oscillation modes. In Case #1, the studied generator contributes negative inter-area damping due to the high excitation gain. In Case #2, the decrease of the excitation gain and the installation of a PSS result in positive damping contribution. The damping ratio of Case #3 falls between the first two cases, indicating that wind penetration can either enhance or undermine the inter-area damping ratio. This depends on the overall contribution of replaced synchronous generators, as WTGs only lightly react to inter-area oscillations. The conclusions from the two-area system are consistent with that of the detailed EI model but with more discernable difference in simulation.

978-1-5090-4168-8/16/$31.00 ©2016 IEEE




| Case | Oscillation Frequency | Oscillation Damping | Oscillation Magnitude |
|---|---|---|---|
| #1: High-gain exciter | 0.79 Hz | 4.76 % | 0.0018 |
| #2: Low-gain exciter with PSS | 0.81 Hz | -1.16 % | 0.0060 |
| #3: WTG | 0.94 Hz | 3.40 % | 0.0340 |

Figure 10. Inter-area Oscillation Analysis of the Two-Area System

## IV. CONCLUSIONS

To assess the impact of wind penetration on rotor angle stability and inter-area oscillation in the EI system, this study looked at critical clearing time and inter-area oscillation damping. From the comparison between the baseline and high-wind cases, dynamic simulation results indicate the following:

- At the 17% penetration level, wind generation has no impact on rotor angle stability for synchronous generators distant to wind generation regions. Nearby generators may be slightly affected with improved stability margin. The improvement is likely linked to the fact that WTGs can provide faster and stronger reactive power support during the first swing and raise power transfer capability.

- WTGs are only lightly engaged in oscillations because of their strict output power regulation. Depending on the damping contribution of displaced synchronous units, wind penetration can either increase or decrease inter-area oscillation damping, but it is certain that wind penetration will increase inter-area oscillation frequency due to the reduction of system inertia. For the same reason, inter-area oscillations can exhibit higher oscillation magnitude.

This study on the stability impact of wind penetration is far from comprehensive. Future work can be improved in three directions: First, the coverage of a full spectrum of operating conditions, like spring light and summer peak, will give a wider range of performance variation. Second, simulating a much higher percentage of wind penetration, say 60%, can better identify the risks and benefits introduced by WTGs. Last, a larger simulation pool containing thousands of cases would provide more solid conclusions. Ongoing research will address these issues.

## REFERENCE


[1] R. Wiser and M. Bolinger, "2014 Wind Technologies Market Report," Lawrence Berkeley National Laboratory, Berkeley, California, 2015.

[2] "U.S. Wind Industry Third Quarter 2015 Market Report," American Wind Energy Association, 2015.

[3] "20% Wind Energy by 2030: Increasing Wind Energy's Contribution to U.S. Electricity Supply," The U.S. Department of Energy, July 2008.

[4] E. Vittal, M. O'Malley, and A. Keane, "Rotor Angle Stability With High Penetrations of Wind Generation," *IEEE Transactions on Power Systems,* vol. 27, pp. 353-362, 2012.

[5] M. Edrah, K. L. Lo, and O. Anaya-Lara, "Impacts of High Penetration of DFIG Wind Turbines on Rotor Angle Stability of Power Systems," *IEEE Transactions on Sustainable Energy,* vol. 6, pp. 759-766, 2015.

[6] G. Tsourakis, B. M. Nomikos, and C. D. Vournas, "Contribution of Doubly Fed Wind Generators to Oscillation Damping," *IEEE Transactions on Energy Conversion,* vol. 24, pp. 783-791, 2009.

[7] D. Gautam, V. Vittal, and T. Harbour, "Impact of Increased Penetration of DFIG-Based Wind Turbine Generators on Transient and Small Signal Stability of Power Systems," *IEEE Transactions on Power Systems,* vol. 24, pp. 1426-1434, 2009.

[8] J. Quintero, V. Vittal, G. T. Heydt, and Z. Hui, "The Impact of Increased Penetration of Converter Control-Based Generators on Power System Modes of Oscillation," *IEEE Transactions on Power Systems,* vol. 29, pp. 2248-2256, 2014.

[9] G. Kou, Y. Liu, S. Hadley, and T. King, "Developing dynamic models for the 2030 Eastern Interconnection grid," in *2014 IEEE PES General Meeting Conference & Exposition*, 2014, pp. 1-5.

[10] G. Kou, P. Markham, S. Hadley, T. King, and Y. Liu, "Impact of Governor Deadband on Frequency Response of the U.S. Eastern Interconnection," *IEEE Transactions on Smart Grid,* vol. PP, pp. 1-1, 2015.

[11] G. Kou, M. Till, T. Bilke, S. Hadley, Y. Liu, and T. King, "Primary Frequency Response Adequacy Study on the U.S. Eastern Interconnection (EI) under High Wind Penetration Conditions," *IEEE Power and Energy Technology Systems Journal,* Accepted 2015.

[12] Y. Liu, L. Zhan, Y. Zhang, P. N. Markham, D. Zhou, J. Guo, Y. Lei, G. Kou, W. Yao, and J. Chai, "Wide-Area Measurement System Development at the Distribution Level: an FNET/GridEye Example," *IEEE Transactions on Power Delivery,* vol. PP, pp. 1-1, 2015.

[13] S. Stavrinos, A. G. Petoussis, A. L. Theophanous, S. Pillutla, and F. S. Prabhakara, "Development of a validated dynamic model of Cyprus Transmission system," in *7th Mediterranean Conference and Exhibition on Power Generation, Transmission, Distribution and Energy Conversion (MedPower 2010)*, 2010, pp. 1-11.

[14] J. C. Mantzaris, A. Metsiou, and C. D. Vournas, "Analysis of Interarea Oscillations Including Governor Effects and Stabilizer Design in South-Eastern Europe," *IEEE Transactions on Power Systems,* vol. 28, pp. 4948-4956, 2013.

[15] K. Clark, N. Miller, and J. Sanchez-Gasca, "Modeling of GE Wind Turbine-Generators for Grid Studies," General Electric International, Inc., 2010.

[16] P. Pourbeik, "Specification of the Second Generation Generic Models for Wind Turbine Generators," Western Electricity Coordinating Council, 2014.

[17] P. Kundur, J. Paserba, V. Ajjarapu, G. Andersson, A. Bose, C. Canizares, N. Hatziargyriou, D. Hill, A. Stankovic, C. Taylor, T. Van Cutsem, and V. Vittal, "Definition and classification of power system stability IEEE/CIGRE joint task force on stability terms and definitions," *IEEE Transactions on Power Systems,* vol. 19, pp. 1387-1401, 2004.

[18] M. L. Crow and A. Singh, "The matrix pencil for power system modal extraction,", *IEEE Transactions on Power Systems,* vol. 20, pp. 501-502, 2005.

[19] P. Kundur, *Power System Stability and Control*. New York, NY, USA: McGraw-Hill, 1994.


978-1-5090-4168-8/16/$31.00 ©2016 IEEE